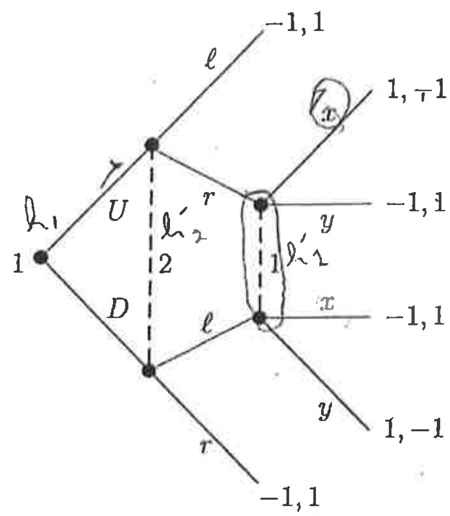
Final

Instructions. The exam contains two questions, each with several parts.

1. Consider the following game in extensive form.



UX DY

a. Define the set of pure, mixed, and a behavior strategies for player 1.

b. Is this a game of perfect recall? Briefly explain.

c. Consider player 1's pure strategies $[U, x]$ and $[D, y]$. Let μ be player 1's mixed strategy in which each of the above pure strategies is played with probability 1/2.

Does player 1 have a behavior strategy to replace μ such that will give player 1 the same payoff against every strategy of player 2? If so provide the behavior strategy. Please explain your answer briefly? → NO, BECAUSE NO PERFECT RECALL

$1/3$ $1/3$ $1/3$

2. Consider the following game. Nature determines $t \in \{2, 3, 4\}$ with equal probability. Player 1 observes the realization of t and chooses $e \in \{0, 1, 2\}$. Player 2 observes e (but does not observe t) and chooses $w \in \mathbb{R}$. Payoff functions are → signaling

$$u_1(t, e, w) = w - \frac{ce}{t^2} \rightarrow \text{worker}$$

$$u_2(t, e, w) = -(w - t)^2$$

(*) CHECK IC

where $c > 0$ is a constant. → only that he does not want to be me up → all choose

a. Determine the values of c, if any, for which there is a pooling WPBE where type $t = 3$ chooses $e = 0$. → all choose e

b. Determine the values of c, if any, for which there is a separating WPBE where type $t = 2$ chooses $e = 0$ and type $t = 3$ chooses $e = 1$.

c. If you found a pooling equilibrium, is that pooling equilibrium compatible with the notion of forward induction? Explain briefly. ⇒ NO!! IF SEES DEVIATION NEEDS TO BELIVE HIGH